

1-0 TOOLS AND EQUIPMENT

SPECIAL TOOLS			
ITEM	PART #	DESCRIPTION	QTY
1	46-287399P1	Decoupling Test Phantom	1
2	46-194427P49	Beckman 3030 DVM	1
3	46-255836P40	Vector Impedance Meter	1

FIELD SUPPLIED TOOLS			
ITEM		DESCRIPTION	QTY
1		15 mm open end wrench	1
2		3/16 inch flat blade screwdriver	1
3		5/16 inch open end wrench	1
4		Utility knife	1
5		6 foot ruler	1

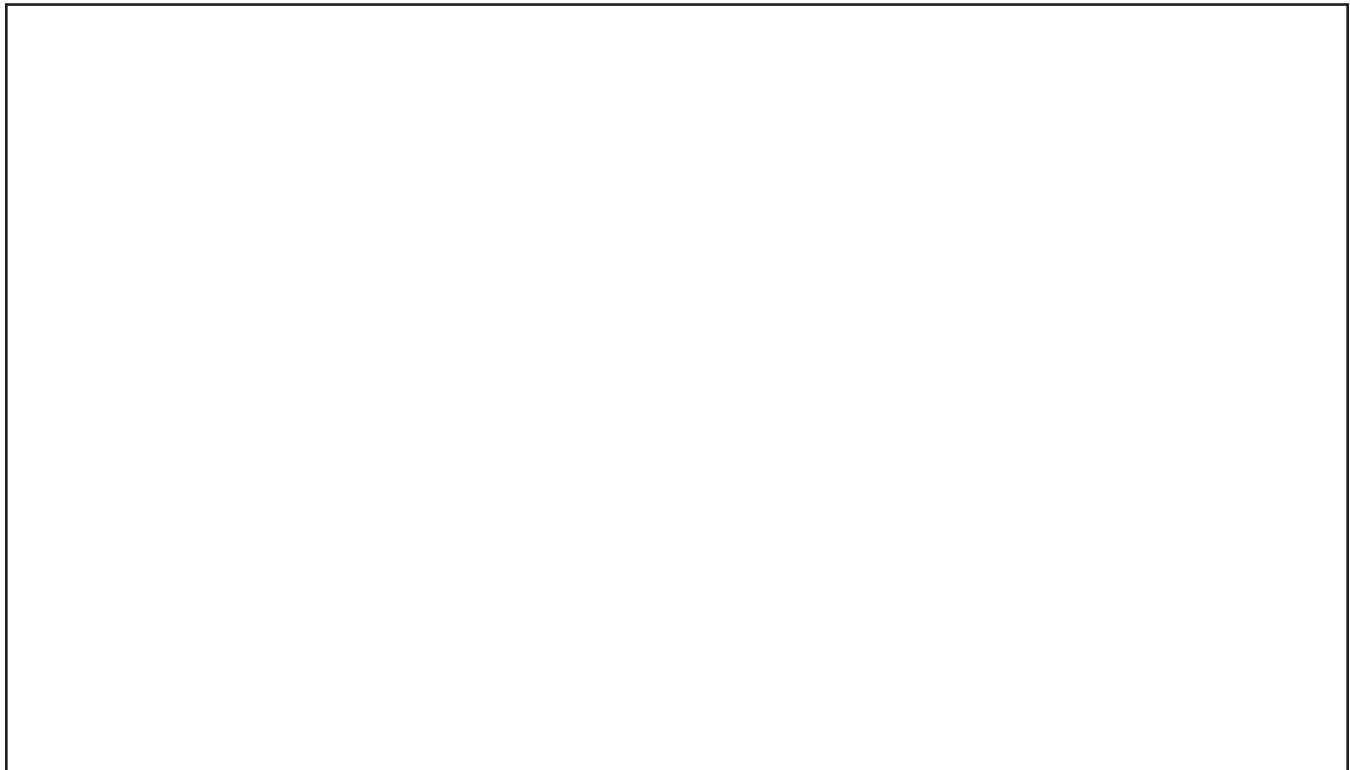
2-0 SIGNA® ADVANTAGE™ 0.5T SCANNER VERIFICATION

- 2-1 Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling.

NOTE

The surface coil decoupling phantom must be filled with distilled water and the supplied cupric sulfate prior to use for any imaging tests. To properly prepare the phantom, completely fill the one liter Nalgene bottle supplied with the coil with distilled water while mixing in the vial of cupric sulfate. Ensure that all of the cupric sulfate has dissolved, and that there are no air bubbles when the cap is replaced on the bottle.

- 2-2 Select **[NEW EXAM]** to enable the patient table **LANDMARK** function. Locate the surface coil decoupling phantom on the patient table and position as shown in ILLUSTRATION 2-1. Do not place the Shoulder Coil under the decoupling phantom. Raise the decoupling phantom to the same approximate location as the surface coil. **LANDMARK** on the center of the decoupling phantom.
- 2-3 Set up the scan on the SIGNA ADVANTAGE 0.5T system using the parameters in ILLUSTRATION 2-2. Perform an **AUTO PRESCAN** to calibrate the RF power level. Record the value for TG for future use in Section 3.
- 2-4 Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.
- 2-5 For the SNR comparisons, select **[CANCEL]**, **[NEW SERIES]**, and acquire two identical images using the scan parameters in ILLUSTRATION 2-3. After acquiring the first image, select **SCAN** to obtain a second. Note the series number for future reference.



DECOUPLING PHANTOM PLACEMENT

ILLUSTRATION 2-1

<u>PATIENT STUDY PARAMETERS</u>		<u>SCANNING RANGE</u>	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
		End Location:	0
<u>PATIENT POSITION</u>		No of Scan Locations	1
Patient Entry:	[HEAD FIRST]	FOV Center (L/R)	0 (P/A) 0
Patient Position:	[SUPINE]		[<] [ACQUISITION TIME]
Axial/Sag Landmark:	[STERNAL NOTCH]	<u>ACQUISITION TIME</u>	
Coil Type:	[BODY]	Acq. Matrix:	[256 x 128]
Scan Plane:	[AXIAL]	Frequency Direction:	[R/L]
	[IMAGING PARAMETERS]	Imaging Time:	[1 Nex]
		Contrast:	[NO]
<u>IMAGING PARAMETERS</u>		Table Delta	0
Image mode:	[2D]		[SCAN SETUP]
Pulse Sequence:	[SPIN ECHO]	<u>SCAN SETUP</u>	
Imaging Options:	[NONE]	Auto CF:	[WATER]
Enter PSD Filename:		Receive B/W (+/-):	[16 KHz]
Monitor SAR?:	Y	Receive B/W (echos 2-4):	[16 KHz]
	[NEXT SCREEN]	Primary Arch. Node:	Default
	[SCAN TIMING]		[REVIEW SCREEN]
<u>SCAN TIMING</u>		<u>REVIEW</u>	
Number of Echos:	[4]		[SCAN OPS]
Echo Time (TE):	[20 ms]	<u>SCAN OPERATIONS</u>	
Rep Time (TR):	[OTHER] [600 ms]		[AUTO PRESCAN]
	[SCANNING RANGE]		[SCAN]

SYSTEM TEST SCAN PARAMETERS

ILLUSTRATION 2-2

<u>PATIENT STUDY PARAMETERS</u>		<u>SCANNING RANGE</u>	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
		End Location:	0
		No of Scan Locations	1
		FOV Center (L/R)	0 (P/A) 0
<u>PATIENT POSITION</u>			[<] [ACQUISITION TIME]
Patient Entry:	[FEET FIRST]	<u>ACQUISITION TIME</u>	
Patient Position:	[SUPINE]	Acq. Matrix:	[256 x 128]
Axial/Sag Landmark:	[STERNAL NOTCH]	Frequency Direction:	[R/L]
Coil Type:	[BODY]	Imaging Time:	[1 Nex]
Scan Plane:	[AXIAL]	Contrast:	[NO]
	[IMAGING PARAMETERS]	Table Delta	0
<u>IMAGING PARAMETERS</u>			[SCAN SETUP]
Image mode:	[2D]	<u>SCAN SETUP</u>	
Pulse Sequence:	[SPIN ECHO]	Auto CF:	[WATER]
Imaging Options:	[NONE]	Receive B/W (+/-):	[16 KHz]
Enter PSD Filename:		Primary Arch. Node:	Default
Monitor SAR?:	Y		[REVIEW SCREEN]
	[NEXT SCREEN]		
	[SCAN TIMING]	<u>REVIEW</u>	
<u>SCAN TIMING</u>			[SCAN OPS]
Number of Echos:	[1]	<u>SCAN OPERATIONS</u>	
Echo Time (TE):	[20 ms]		
Rep Time (TR):	[OTHER] [600 ms]		[AUTO PRESCAN]
	[SCANNING RANGE]		[SCAN]

BODY COIL SNR SCAN PARAMETERS

ILLUSTRATION 2-3

3-0 SHOULDER COIL IMAGING PERFORMANCE VERIFICATION

- 3-1 Set up the SIGNA ADVANTAGE 0.5T Shoulder coil and decoupling phantom on the patient table as shown in ILLUSTRATION 2-1. Connect the surface coil cable to the **SURFACE COIL QUICK DISCONNECT BOX** connected to the **HEAD COIL PORT**.
- 3-2 Select [**CANCEL**], [**NEW SERIES**], and repeat the scan using the parameters in ILLUSTRATION 3-1. Use **AUTO PRESCAN** to set the values of **R1** and **R2**. If the value of TG does not match the value obtained in step 2-3, use **MANUAL PRESCAN** to set the value of TG to the value found in step 2-3.
- 3-3 Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images. Study the third and fourth echoes using a **WINDOW WIDTH** and **WINDOW LEVEL** to obtain good contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, see ILLUSTRATION 3-2. A decoupling failure will cause holes, distortions,

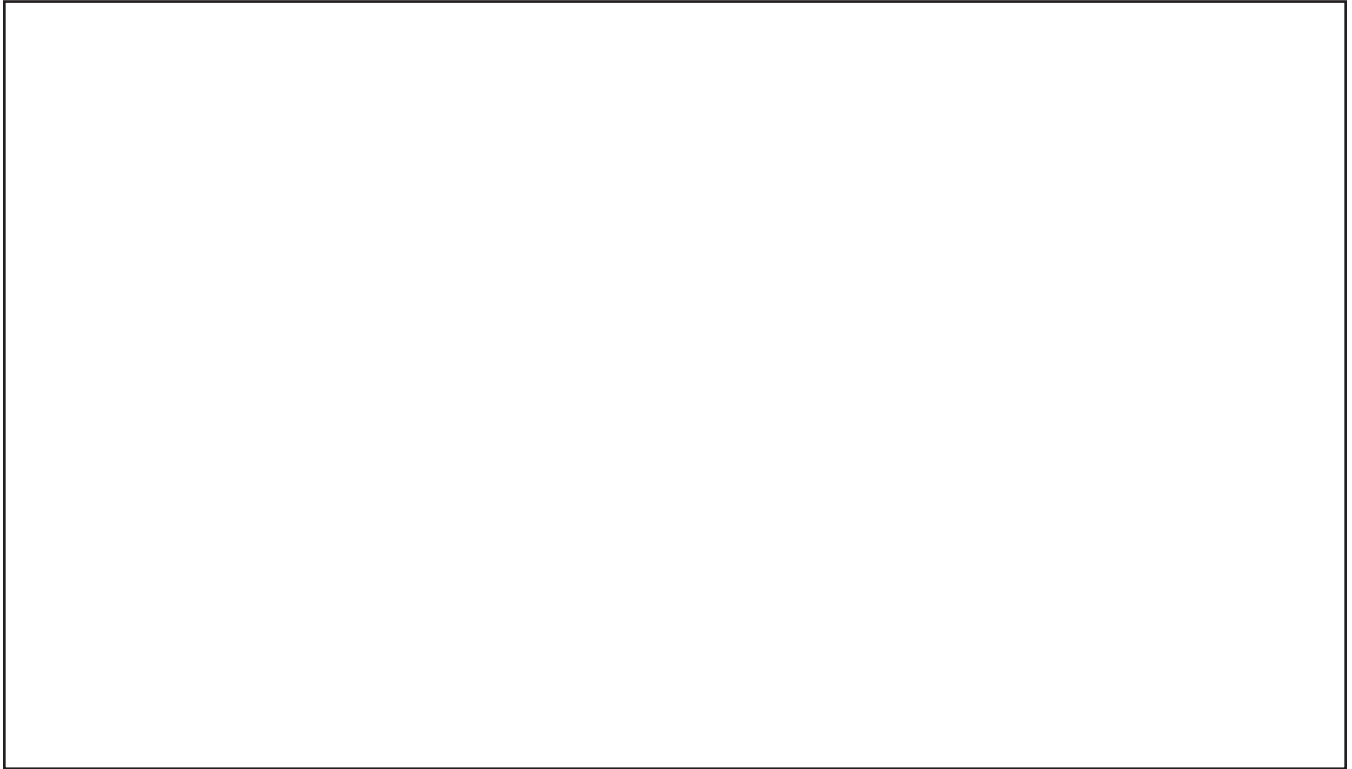
<u>PATIENT STUDY PARAMETERS</u>		<u>SCANNING RANGE</u>	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
		End Location:	0
		No of Scan Locations	1
		FOV Center (L/R)	0 (P/A) 0
<u>PATIENT POSITION</u>			[<] [ACQUISITION TIME]
Patient Entry:	[HEAD FIRST]	<u>ACQUISITION TIME</u>	
Patient Position:	[SUPINE]	Acq. Matrix:	[256 x 128]
Axial/Sag Landmark:	[STERNAL NOTCH]	Frequency Direction:	[R/L]
Coil Type:	[OTHER] [SHOULDER]	Imaging Time:	[1 Nex]
Scan Plane:	[SAGITTAL]	Contrast:	[NO]
	[IMAGING PARAMETERS]	Table Delta	0
<u>IMAGING PARAMETERS</u>			[SCAN SETUP]
Image mode:	[2D]	<u>SCAN SETUP</u>	
Pulse Sequence:	[SPIN ECHO]	Auto CF:	[WATER]
Imaging Options:	[NONE]	Receive B/W (+/-):	[16 KHz]
Enter PSD Filename:		Receive B/W (echos 2-4):	[16 KHz]
Monitor SAR?:	Y	Primary Arch. Node:	Default
	[NEXT SCREEN]		[REVIEW SCREEN]
	[SCAN TIMING]		
<u>SCAN TIMING</u>		<u>REVIEW</u>	[SCAN OPS]
Number of Echos:	[4]		
Echo Time (TE):	[20 ms]	<u>SCAN OPERATIONS</u>	
Rep Time (TR):	[OTHER] [600 ms]		
	[SCANNING RANGE]		[AUTO PRESCAN]
			[SCAN]

SURFACE COIL DECOUPLING TEST SCAN PARAMETERS

ILLUSTRATION 3-1

or striped patterns in the image as seen in ILLUSTRATION 3-3. Refer to Section 4-0 to troubleshoot decoupling failures.

- 3-4 Select **[CANCEL]**, **[NEW SERIES]**, and acquire two identical images using the scan parameters in ILLUSTRATION 3-4. After acquiring the first image, select **SCAN** to obtain a second.
- 3-5 Compare the signal to noise ratio of the Shoulder Coil image of step 3-4 with the signal to noise ratio of the body coil image of step 2-5. See Section 4-0, *SNR Image Analysis*. The signal to noise ratio of the Shoulder Coil should be 2.5 times greater than the body coil signal to noise ratio.
- 3-6 Inspect the images for visible ghosting [similar in appearance to motion artifacts] in the phase encoding direction. If images contain ghosting artifacts, an intermittent cable or connection is suspected. Refer to Section 5-0 for troubleshooting information.



NORMAL DECOUPLING

ILLUSTRATION 3-2



DECOUPLING FAILURE

ILLUSTRATION 3-3

<u>PATIENT STUDY PARAMETERS</u>		<u>SCANNING RANGE</u>	
ID:	GESERVICE	Field of View:	[32]
Name:		Slice Thickness:	[5 mm]
Patient Weight:	111 LBS	Interscan Spacing:	[0]
	[PATIENT POSITION]	Start Location:	0
		End Location:	0
		No of Scan Locations:	1
		FOV Center (L/R):	0 (P/A) 0
<u>PATIENT POSITION</u>			[<] [ACQUISITION TIME]
Patient Entry:	[HEAD FIRST]	<u>ACQUISITION TIME</u>	
Patient Position:	[SUPINE]	Acq. Matrix:	[256 x 128]
Axial/Sag Landmark:	[STERNAL NOTCH]	Frequency Direction:	[R/L]
Coil Type:	[OTHER] [SHOULDER]	Imaging Time:	[1 Nex]
Scan Plane:	[AXIAL]	Contrast:	[NO]
	[IMAGING PARAMETERS]	Table Delta:	0
<u>IMAGING PARAMETERS</u>			[SCAN SETUP]
Image mode:	[2D]	<u>SCAN SETUP</u>	
Pulse Sequence:	[SPIN ECHO]	Auto CF:	[WATER]
Imaging Options:	[NONE]	Receive B/W (+/-):	[16 KHz]
Enter PSD Filename:		Primary Arch. Node:	Default
Monitor SAR?:	Y		[REVIEW SCREEN]
	[NEXT SCREEN]		
	[SCAN TIMING]	<u>REVIEW</u>	
<u>SCAN TIMING</u>			[SCAN OPS]
Number of Echos:	[1]	<u>SCAN OPERATIONS</u>	
Echo Time (TE):	[20 ms]		
Rep Time (TR):	[OTHER] [600 ms]		
	[SCANNING RANGE]		[AUTO PRESCAN]
			[SCAN]

SHOULDER COIL SNR SCAN PARAMETERS

ILLUSTRATION 3-4

4-0 SNR IMAGE ANALYSIS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

4-1 Procedure

- 4-1-1 Touch [UTILITIES], [MRTools], [Image Quality], then [SNRTest]. The SNR Test screen is displayed. See ILLUSTRATION 4-1.
- 4-1-2 Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select [List Exams], [List Series], or [List Images] to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

- 4-1-3 Enter second image series and image numbers. If series or image numbers are not known, select [**List Series**] or [**List Images**] to display list. The second image Exam Number is defaulted to the same as first image.
- 4-1-4 Select [**Filter Off**] and [**Image Off**].
- 4-1-5 Touch [**Accept**] to begin analysis. The final values are displayed on the screen. See ILLUSTRATION 4-1.
- 4-1-6 Record Signal to Noise Ratio on ILLUSTRATION 4-2.
- 4-1-7 Touch [**Continue**] then select the next exam and repeat the above analysis for each image pair.

NEMA STANDARD SNR TEST

Filter
Off

Image
Off

First
Image

List/
Sel
Exams

List/
Sel
Series

List/
Sel
Images

Second
Image

List/
Sel
Series

List/
Sel
Images

Body Axial SNR Data

Signal : xx.x

Noise : xx.x

SNR : xx.x

Cancel

Utility
Main

MR
Tools
Main

Image
Qual-
ity

Accept

 Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN

ILLUSTRATION 4-1

Coil	Signal to Noise Ratio	Date
Body Coil		
Shoulder Coil		

SNR VALUES

ILLUSTRATION 4-2

5-0 TEST AND TROUBLESHOOTING PROCEDURES

5-1 Coil Impedance Measurements

- 5-1-1 Move to an area where the magnetic field from the SIGNA ADVANTAGE 0.5T magnet is less than 5 Gauss. Load the surface coil with a human subject. Use the procedure in the OPERATION MANUAL to place the surface coil.
- 5-1-2 Connect the surface coil cable to the vector impedance meter probe using an appropriate adapter. Use the vector impedance meter to measure the coil impedance at 21.29 MHz. Ensure that the cable does not touch or pass close to the coil housing while making this measurement.
- 5-1-3 The impedance magnitude should nominally be 50 ohms under loaded conditions. It should fall within the range of 37.5 to 62.5 ohms under test conditions.
- 5-1-4 The phase angle should nominally be zero degrees. It should fall within the range of -15 to +15 degrees under test conditions.

5-2 Coil Integrity Tests

- 5-2-1 Repeat steps 5-1-1 to 5-1-4 while flexing the surface coil cable, especially at the fitting and the coil housing strain relief. Using the **HIGH SPEED** mode on the vector impedance meter, note any erratic fluctuations in the coil impedance. Any erratic or unstable readings may indicate a defective cable or connector. Refer to Section 6-0 for service procedures.

5-3 Decoupling Diode Tests

- 5-3-1 Use the digital multimeter on the diode test range to measure across the coil connector. With the negative lead to the connector shell and the positive to the connector center pin, the reading should be between 0.5 and 1.0. Reversing the leads should cause the reading to be infinity [overrange].
- 5-3-2 An open reading in both directions indicates an open diode, or an open cable. Refer to Section 5-4 to check the cable. Refer to Section 6-4 to replace the diode.
- 5-3-3 A short circuit or excessively low reading indicates a shorted diode or a shorted cable. Refer to Section 5-4 to check the cable. Refer to Section 6-4 to replace the diode.

5-4 Cable Tests

- 5-4-1 If the surface coil passes the tests in Sections 5-1, 5-2, and 5-3, the cable can be assumed to be functional.
- 5-4-2 If a short or open circuit was discovered in Section 5-3, the cable may be tested by disconnecting the cable at the PC board via the service access cover, as shown in ILLUSTRATION 5-1. The cable may then be tested for continuity of the center conductor and shield, and for shorts between the center conductor and shield using the multimeter continuity test function.

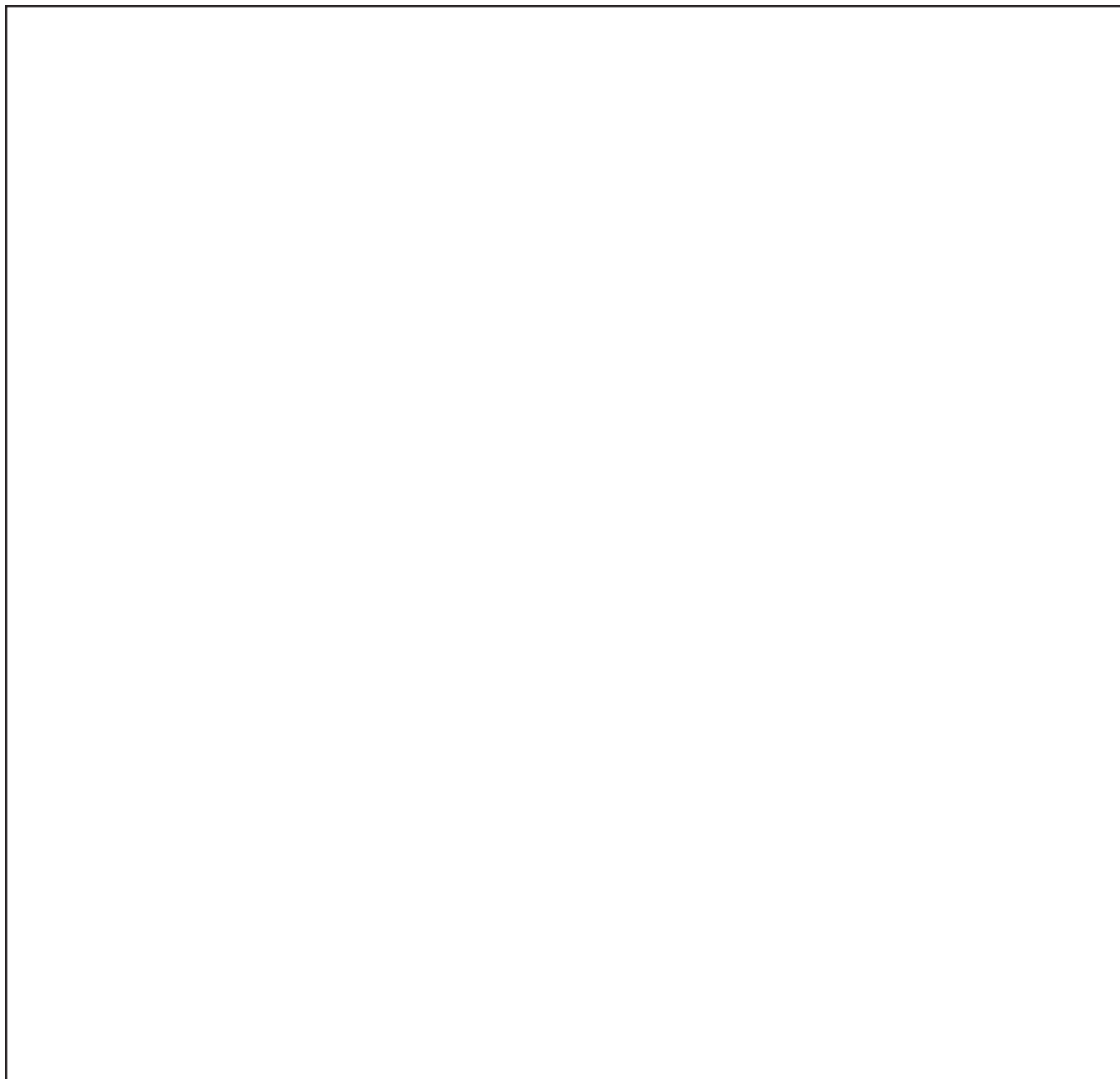
**ACCESS COVER, INTERNAL CABLE CONNECTION**

ILLUSTRATION 5-1

6-0 SERVICE PROCEDURES

- 6-1 A service kit is available for replacement of the cable assembly. ILLUSTRATION 6-1 describes the contents of the kit.

CABLE KIT 46-317822P1 — 30523C25001 - Consisting of the following items:			
ITEM	VENDOR PART#	DESCRIPTION	QTY
1	30511C25001	Cable Assembly	1
2	87305-T-165	Service Manual - SIGNA 0.5T	1

CABLE ASSEMBLY SERVICE KIT

ILLUSTRATION 6-1

6-2 Coaxial Cable Replacement - ILLUSTRATION 6-2

- 6-2-1 Remove the service access cover from the housing by removing the four nylon screws attaching the cover.
- 6-2-2 Disconnect the coaxial cable by unscrewing the cable SMA connector from the SMA connector on the PC board.
- 6-2-3 Loosen and remove the strain relief body from the strain relief base using a 15 MM open end wrench. Slide the strain relief body down the cable away from the base.
- 6-2-4 Loosen and remove the retaining nut from the strain relief base using a 3/4 inch open end wrench.
- 6-2-5 Remove the cable assembly by removing the strain relief base from the hole in the housing of the coil.
- 6-2-6 Remove the retaining nut from the new cable assembly. Insert the SMA end of the cable through the strain relief mounting hole in the coil bottom cover. Place the retaining nut over the SMA end of the cable.
- 6-2-7 Reconnect the cable SMA connector to the SMA connector on the PC board. Tighten firmly.
- 6-2-8 Attach the strain relief base by inserting it into the hole in the housing of the coil. Reinstall and tighten the retaining nut using a 3/4 inch wrench. Tighten the nut one-half turn beyond initial contact with the surface of the housing.
- 6-2-9 Using a 15 MM wrench, reinstall and tighten the strain relief body to the base until the base begins to turn [8 inch-pounds].
- 6-2-10 Attach the service access cover to the coil housing using the four nylon screws.
- 6-2-11 Repeat the tests in Sections 3-0 and 4-0 to verify proper coil operation.

6-3 Output Connector Replacement - ILLUSTRATION 6-2

- 6-3-1 If the cable will be shortened to less than 60-1/8 inches by a repair, it is recommended that the entire cable be replaced in the event of a connector or connector-end cable failure.
- 6-3-2 Replace the connector using ILLUSTRATION 6-2 as a guide. Use a new Tri-Lug BNC coaxial fitting - part number 46-317498P1, and crimp tool - part number 46-255841, with insert - part number 46-255841P100, to install the replacement connector on the cable.
- 6-3-3 Repeat the tests in Sections 3-0 and 4-0 to verify proper coil operation.

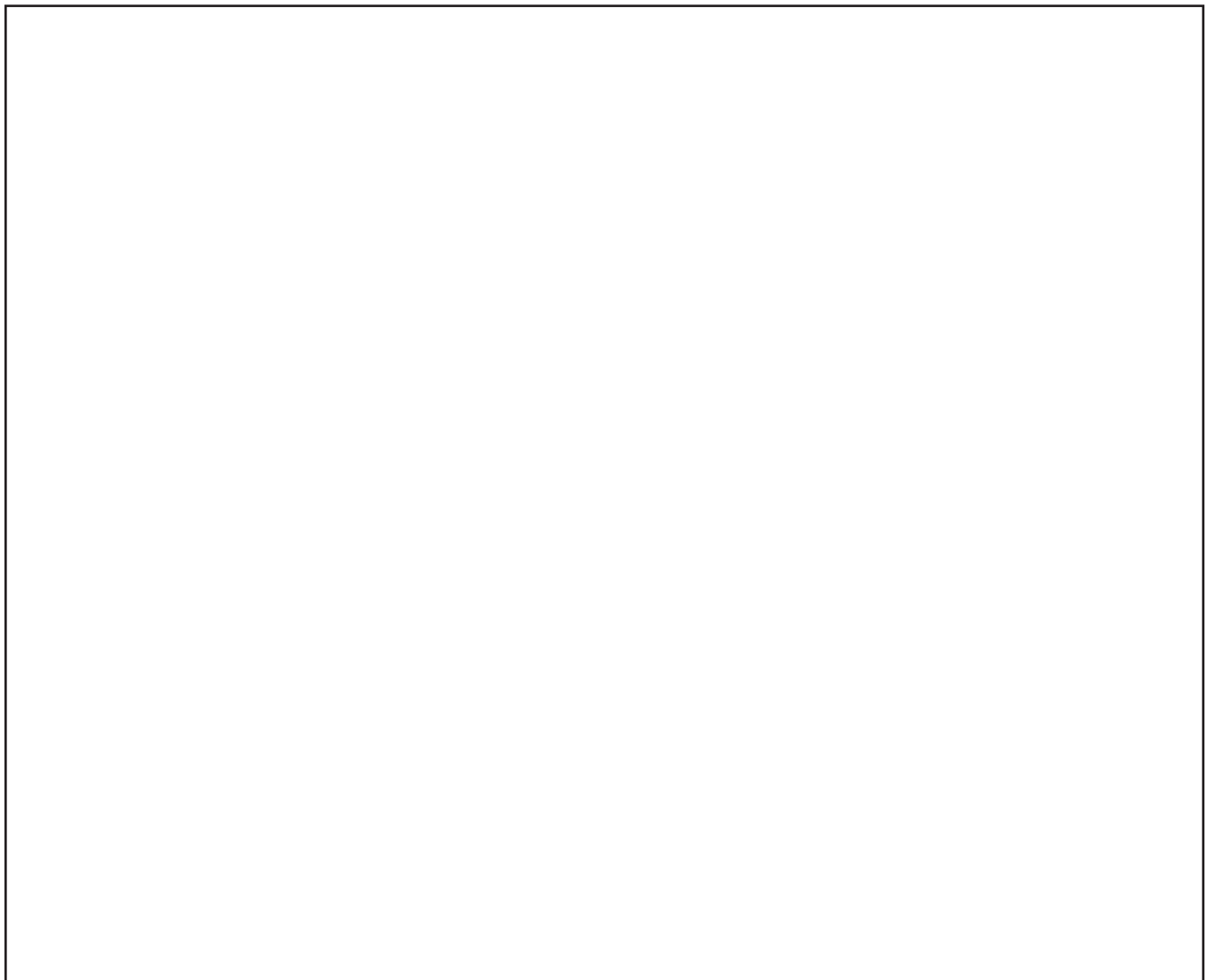
**OUTPUT CONNECTOR/COAXIAL CABLE REPLACEMENT**

ILLUSTRATION 6-2

6-4 Diode Replacement

6-4-1 If it has been determined that a diode must be replaced - Section 5-3, remove the access cover.

6-4-2 Diode placement is shown in ILLUSTRATION 6-3, with the schematic shown in ILLUSTRATION 6-4.

6-4-3 Unsolder defective diode and replace with new diode - part number 46-320075P1.

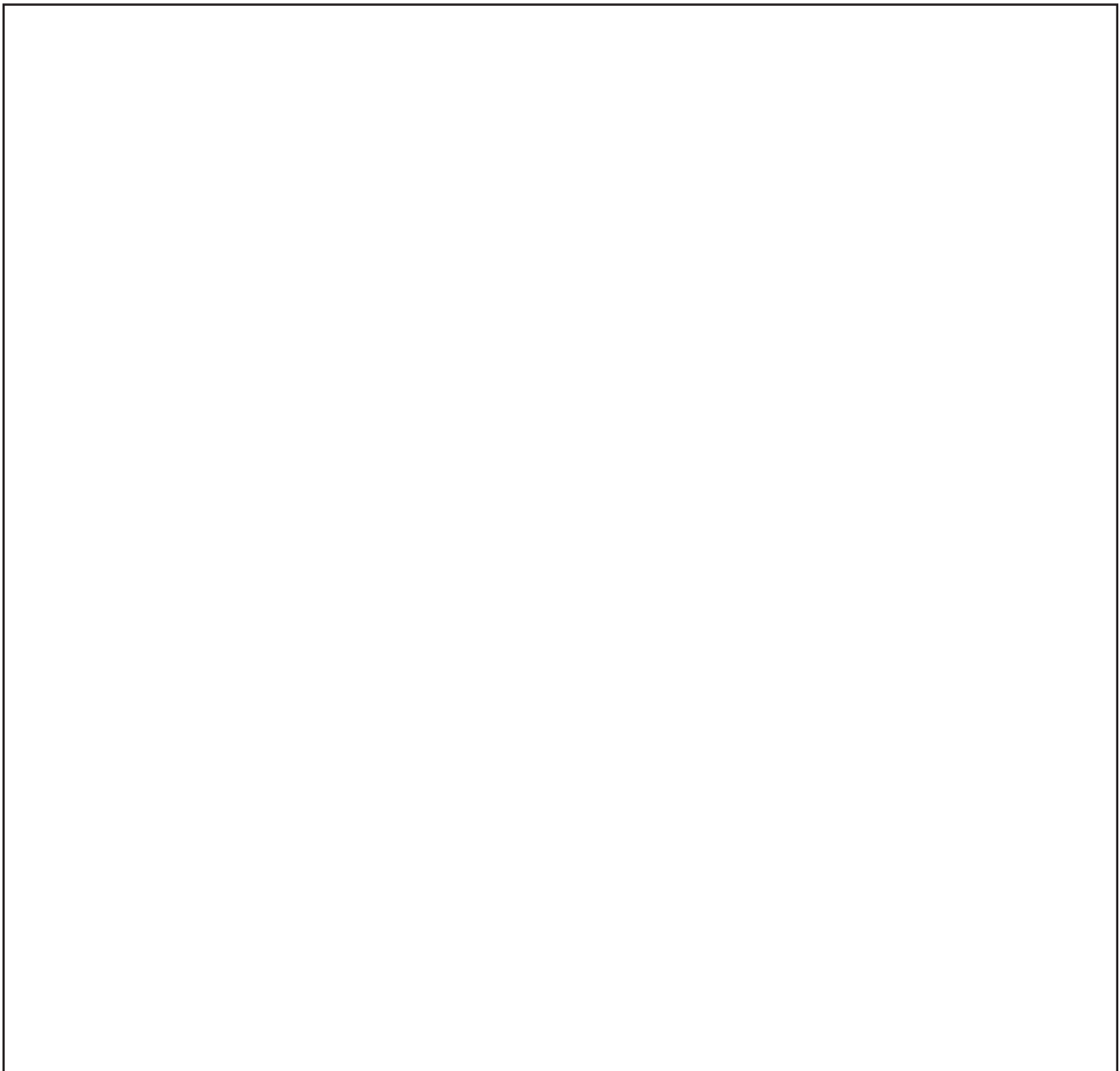
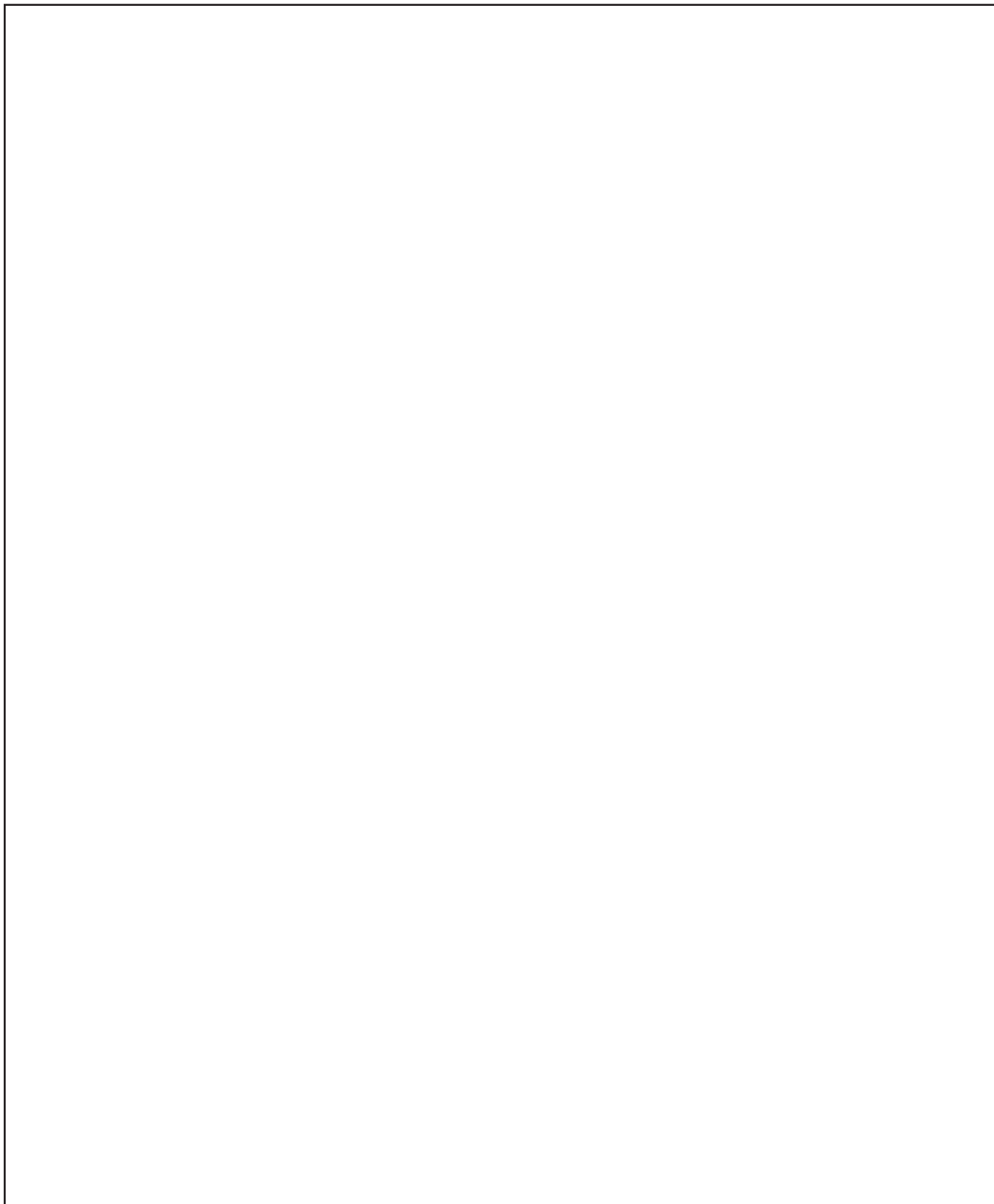
**DIODE LOCATION**

ILLUSTRATION 6-3



SHOULDER COIL SCHEMATIC

ILLUSTRATION 6-4

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: _____

SITE NAME: _____

SITE ADDRESS: _____

SERVICE ENGINEER: _____

COIL SERIAL NUMBER: _____

DATE COIL INSTALLED: _____

DESCRIPTION OF COIL PROBLEM:

ELECTRICAL CHECKS	
VECTOR IMPEDANCE METER TEST - COIL LOADED WITH HUMAN SUBJECT IN FREE SPACE	
MAGNITUDE AT 21.29 MHz	[50 OHMS]
PHASE AT 21.29 MHz	[0°]
VECTOR IMPEDANCE METER TEST - COIL UNLOADED IN FREE SPACE	
MAGNITUDE AT 21.29 MHz	
PHASE AT 21.29 MHz	